

# STAT452

## Multiple Linear Regression

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# Statistical Models

A Statistical model is a simple, low-dimensional (as few predictors as possible) summary of

- the relationship present in the data
- the data-generation process
- the relationship present in the population

Statistical models allow **errors (uncertainty)**

$$\begin{array}{ccccc} Y & = & f(X_1, X_2, \dots, X_p) & + & \varepsilon \\ \text{response} & & \text{deterministic function} & & \text{error (noise)} \end{array}$$

# Linear regression models

We will focus on *linear regression models* where

$$Y = f(X_1, X_2, \dots, X_p) + \varepsilon = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \varepsilon$$

The adjective *linear* means the model is linear in its parameters  $\beta_0, \beta_1, \dots, \beta_p$ . For example, the following are linear regression models:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon,$$

$$Y = \beta_0 + \beta_1 \log(X) + \varepsilon,$$

even though the relationship between  $Y$  and  $X$  is not linear.

# Linear regression models

Here are some non-linear models can be turned linear

## Example 1.

Non-linear model:

$$Y = \frac{X}{\alpha X + \beta}$$

reciprocal

$$\frac{1}{Y} = \alpha + \beta \left( \frac{1}{X} \right)$$

$$\Downarrow \quad \Downarrow \quad \Downarrow$$

Linear model:

$$Y' = \alpha + \beta X'$$

where  $Y' = 1/Y$ ,  $X' = 1/X$ .

# Linear regression models

## Example 2.

Which of the following models are linear? And which one can be turned linear after transformation?

- a)  $Y = \beta_0 + \beta_1^X + \varepsilon$
- b)  $Y = \beta_0 \beta_1^X \varepsilon$
- c)  $Y = \beta_0 + \beta_1 e^X + \varepsilon$
- d)  $Y = \beta_0 + \beta_1 X^2 + \beta_2 \log(X) + \varepsilon$

# Multiple linear regression models

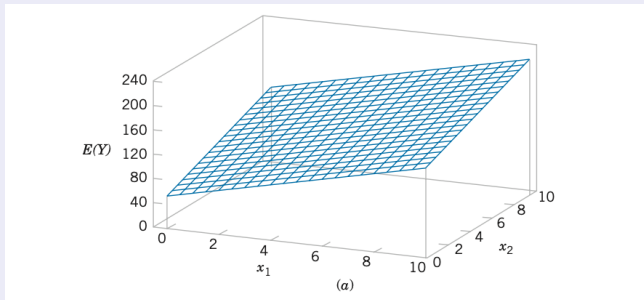
- In STAT451, we studied about simple linear regression (SLR): one predictor variable  $X$ .
- Multiple linear regression: multiple predictor variables  $X_1, X_2, \dots, X_p$ .
- Example:
  - SLR: property tax =  $\beta_0 + \beta_1 \times \text{house price}$ .
  - MLR: property tax =  $\beta_0 + \beta_1 \times \text{house price} + \beta_2 \times \text{house size}$ .
- Question: how to fit multiple linear regression model?

## Multiple linear regression models

### Example 3.

Multiple linear regression model with two regressors (predictor variables):

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon.$$



$$\mathbb{E}(Y) = 50 + 10X_1 + 7X_2.$$



# Data for Multiple linear regression models

| SLR          |              |
|--------------|--------------|
| $\mathbf{X}$ | $\mathbf{Y}$ |
| $x_1$        | $y_1$        |
| $x_2$        | $y_2$        |
| $\vdots$     | $\vdots$     |
| $x_n$        | $y_n$        |

| MLR      |          |          |          |              |
|----------|----------|----------|----------|--------------|
| $X_1$    | $X_2$    | $\cdots$ | $X_p$    | $\mathbf{Y}$ |
| $x_{11}$ | $x_{12}$ | $\cdots$ | $x_{1p}$ | $y_1$        |
| $x_{21}$ | $x_{22}$ | $\cdots$ | $x_{2p}$ | $y_2$        |
| $\vdots$ | $\vdots$ | $\ddots$ | $\vdots$ | $\vdots$     |
| $x_{n1}$ | $x_{n2}$ | $\cdots$ | $x_{np}$ | $y_n$        |

- For SLR, we observe **pairs** of data values.
- For MLR, we observe **rows** of data values.
- Each row (or pair) is called a **case**, a **record**, or a **data point**.
- $y_i$  is the **response** (or **dependent variable**) of the  $i$ th case.
- There are  $p$  **explanatory variables** (or **predictors**, **covariates**), and  $x_{ik}$  is the value of the explanatory variable  $X_k$  of the  $i$ th case.

# Multiple linear regression models

## Model

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \varepsilon_i.$$

In the model above,

- $\varepsilon_i$ 's (errors, or noise) are i.i.d.  $\mathcal{N}(0, \sigma^2)$ .
- Parameters include:
  - $\beta_0$  = intercept.
  - $\beta_k$  = regression coefficient (slope) for the  $k$ th explanatory variable,  $k = 1, \dots, p$ .
  - $\sigma^2 = \text{Var}(\varepsilon_i)$  = the variance of errors.
- Observed (known):  $y_i, x_{i1}, x_{i2}, \dots, x_{ip}$ .
- Unknown:  $\beta_0, \beta_1, \dots, \beta_p, \sigma^2, \varepsilon_i$ 's.
- Random:  $\varepsilon_i$ 's,  $y_i$ 's.
- Constants (not random):  $\beta_k$ 's,  $\sigma^2, x_{ik}$ 's.

# Multiple linear regression models in Matrix form

$$\underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}}_{\mathbf{Y}_{n \times 1}} = \underbrace{\begin{bmatrix} 1 & x_{11} & x_{21} & \cdots & x_{p1} \\ 1 & x_{12} & x_{22} & \cdots & x_{p2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{1n} & x_{2n} & \cdots & x_{pn} \end{bmatrix}}_{\mathbf{X}_{n \times (p+1)}} \underbrace{\begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix}}_{\boldsymbol{\beta}_{(p+1) \times 1}} + \underbrace{\begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}}_{\boldsymbol{\varepsilon}_{n \times 1}}$$

or

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

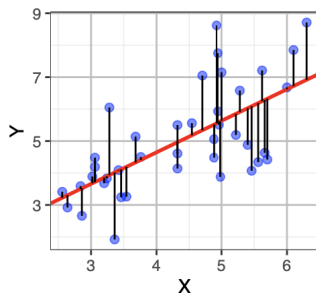
## Fitting the Model — Least Squares Method

Recall for SLR, the least squares estimate  $(\hat{\beta}_0, \hat{\beta}_1)$  for  $(\beta_0, \beta_1)$  is the intercept and slope of the straight line with the minimum sum of squared vertical distances to the data points:

$$\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2.$$

MLR is just like SLR. The least squares estimate  $(\hat{\beta}_0, \dots, \hat{\beta}_p)$  for  $(\beta_0, \dots, \beta_p)$  is the intercept and slopes of the (hyper)plane with the minimum sum of squared vertical distance to the data points:

$$\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_p x_{ip})^2.$$



## Least squares estimate of regression coefficients

To find the  $(\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p)$  that minimize

$$L(\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p) = \sum_{i=1}^n \left( y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_p x_{ip} \right)^2$$

one can set the derivatives of  $L$  with respect to  $\hat{\beta}_j$  to 0:

$$\frac{\partial L}{\partial \hat{\beta}_0} = -2 \sum_{i=1}^n \left( y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_p x_{ip} \right)$$

$$\frac{\partial L}{\partial \hat{\beta}_k} = -2 \sum_{i=1}^n x_{ik} \left( y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_p x_{ip} \right), \quad k = 1, 2, \dots, p$$

and then equate them to 0. This results in a system of  $(p+1)$  equations in  $(p+1)$  unknowns.

## Least squares estimate of regression coefficients

The least squares estimate  $(\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p)$  is the solution to the following system of equations, called the **normal equations**:

$$\begin{aligned}
 \hat{\beta}_0 \cdot n + \hat{\beta}_1 \sum_{i=1}^n x_{i1} + \dots + \hat{\beta}_p \sum_{i=1}^n x_{ip} &= \sum_{i=1}^n y_i \\
 \hat{\beta}_0 \sum_{i=1}^n x_{i1} + \hat{\beta}_1 \sum_{i=1}^n x_{i1}^2 + \dots + \hat{\beta}_p \sum_{i=1}^n x_{i1}x_{ip} &= \sum_{i=1}^n x_{i1}y_i \\
 &\vdots \\
 \hat{\beta}_0 \sum_{i=1}^n x_{ik} + \hat{\beta}_1 \sum_{i=1}^n x_{ik}x_{i1} + \dots + \hat{\beta}_p \sum_{i=1}^n x_{ik}x_{ip} &= \sum_{i=1}^n x_{ik}y_i \\
 &\vdots \\
 \hat{\beta}_0 \sum_{i=1}^n x_{ip} + \hat{\beta}_1 \sum_{i=1}^n x_{ip}x_{i1} + \dots + \hat{\beta}_p \sum_{i=1}^n x_{ip}^2 &= \sum_{i=1}^n x_{ip}y_i
 \end{aligned}$$

## Matrix normal equation

- Least square function:

$$L = \sum_{i=1}^n \varepsilon_i^2 = \varepsilon' \varepsilon = (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta).$$

- Coefficient satisfies:

$$\frac{\partial L}{\partial \beta} = 0.$$

- Normal equation:

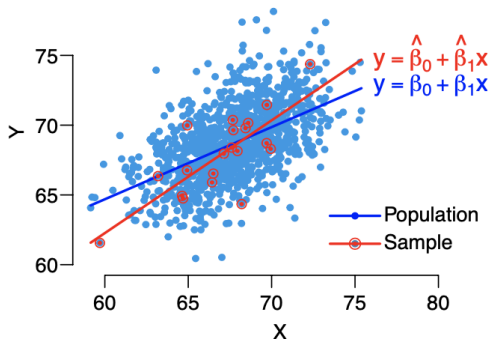
$$\mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{X}'\mathbf{y} \quad \Rightarrow \quad \hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

$$\begin{bmatrix} n & \sum x_{i1} & \sum x_{i2} & \cdots & \sum x_{ip} \\ \sum x_{i1} & \sum x_{i1}^2 & \sum x_{i1}x_{i2} & \cdots & \sum x_{i1}x_{ip} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sum x_{ip} & \sum x_{ip}x_{i1} & \sum x_{ip}x_{i2} & \cdots & \sum x_{ip}^2 \end{bmatrix} \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_p \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_{i1}y_i \\ \vdots \\ \sum x_{ip}y_i \end{bmatrix}$$

## Parameters v.s. Estimates

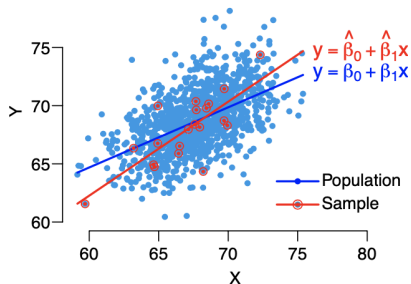
Note that  $\beta_i$ 's are the coefficients of the MLR model, and  $\hat{\beta}_i$ 's are the estimates of  $\beta_i$ 's. For SLR model:

- $y = \beta_0 + \beta_1 x$  is the least square line for the **population**.
- $y = \hat{\beta}_0 + \hat{\beta}_1 x$  is the least square line for a **sample**.





# Population vs. Sample Regression Lines



|                                                          |                                                      |
|----------------------------------------------------------|------------------------------------------------------|
| $y = \beta_0 + \beta_1 x$                                | $y = \hat{\beta}_0 + \hat{\beta}_1 x$                |
| least-square regression line<br>of the <b>population</b> | least-square regression line<br>of the <b>sample</b> |
| fixed                                                    | random, changes<br>from sample to sample             |
| unknown                                                  | can be calculated<br>from sample                     |
| of interest                                              | not of interest                                      |

## Fitted values

The fitted value or predicted value:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \dots + \hat{\beta}_p x_{ip},$$

where

- $\hat{y}_i$  is the fitted value
- $y_i$  is the actual observed value

## Errors and Residuals

- One cannot directly compute the errors

$$\varepsilon_i = y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_p x_{ip}$$

since the coefficients  $\beta_0, \beta_1, \dots, \beta_p$  are **unknown**.

- The errors  $\varepsilon_i$  can be estimated by the **residuals**  $e_i$  defined as

$$\begin{aligned} \text{residual } e_i &= \text{observed } y_i - \text{predicted } y_i \\ &= y_i - \hat{y}_i \\ &= y_i - \underbrace{(\hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \dots + \hat{\beta}_p x_{ip})}_{\text{predicted } y_i}. \end{aligned}$$

- $e_i \neq \varepsilon_i$  in general since  $\hat{\beta}_j \neq \beta_j$ .

## Properties of Residuals

Recall that the LS estimate  $(\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p)$  satisfies the following equations

$$\sum_{i=1}^n \underbrace{(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_p x_{ip})}_{y_i - \hat{y}_i = e_i = \text{residual}} = 0,$$

$$\sum_{i=1}^n x_{ik} \underbrace{(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_p x_{ip})}_{=0} = 0, \quad k = 1, 2, \dots, p.$$

The residuals  $e_i$  hence have these properties

$$\underbrace{\sum_{i=1}^n e_i}_{\text{Residuals add up to 0}} = 0, \quad \underbrace{\sum_{i=1}^n x_{ik} e_i}_{\text{Residuals are orthogonal to predictors}} = 0, \quad k = 1, 2, \dots, p.$$

The two properties combined imply that **the residuals have 0 correlation with each of the  $p$  predictors**, since:

$$\text{Cov}(X_k, e) = \frac{1}{n-1} \left( \underbrace{\sum_{i=1}^n x_{ik} e_i}_{=0} - n \bar{x}_k \underbrace{\bar{e}}_{=0} \right) = 0.$$

## Mean Square Error (MSE) — Estimate of $\sigma^2$

The variance  $\sigma^2$  of the errors  $\varepsilon_i$ 's is estimated by the **mean square error (MSE)**, the sum of squares of residuals divided by  $n - p - 1$ .

$$\text{MSE} = \frac{\sum_{i=1}^n e_i^2}{n - p - 1} = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n - p - 1}.$$

Why divided by  $n - p - 1$  instead of by  $n$ ?

We will show that MSE is an unbiased estimator of  $\sigma^2$ .

## Example: Advertising Data Set

```
Advertising <- read.csv("Advertising.csv")
attach(Advertising)
names(Advertising)
```

```
## [1] "TV"          "Radio"       "Newspaper" "Sales"
```

```
head(Advertising)
```

```
##      TV Radio Newspaper Sales
## 1 230.1  37.8      69.2  22.1
## 2  44.5  39.3      45.1  10.4
## 3  17.2  45.9      69.3   9.3
## 4 151.5  41.3      58.5  18.5
## 5 180.8  10.8      58.4  12.9
## 6   8.7  48.9      75.0   7.2
```

## Example: Advertising Data Set

```

model=lm(Sales~TV+Radio+Newspaper)
summary(model)

##
## Call:
## lm(formula = Sales ~ TV + Radio + Newspaper)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8.8277 -0.8908  0.2418  1.1893  2.8292
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.938889   0.311908   9.422  <2e-16 ***
## TV           0.045765   0.001395  32.809  <2e-16 ***
## Radio        0.188530   0.008611  21.893  <2e-16 ***
## Newspaper   -0.001037   0.005871  -0.177    0.86
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.686 on 196 degrees of freedom
## Multiple R-squared:  0.8972, Adjusted R-squared:  0.8956
## F-statistic: 570.3 on 3 and 196 DF,  p-value: < 2.2e-16

```

# Interpretation of Regression Coefficients

Recall hypothesis testing on parameter estimates,

- Linear relationship between **Sales** and **TV**, and **Sales** and **Radio** are significant.
- Linear relationship between **Sales** and **Newspaper** are NOT significant.

*\*Why?*



## Least Squares Estimates Are Unbiased

The LS estimate  $\hat{\beta}$  in matrix notation is

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}.$$

Based on the MLR model in matrix notation  $\mathbf{Y} = \mathbf{X}\beta + \epsilon$ , the expected value of  $\mathbf{Y}$  is

$$\mathbb{E}[\mathbf{Y}] = \mathbb{E}[\mathbf{X}\beta] + \underbrace{\mathbb{E}[\epsilon]}_{=0} = \mathbf{X}\beta.$$

Recall in MLR,  $\mathbf{X}$  are regarded as **fixed numbers**, no randomness.

The expected value of  $\hat{\beta}$  is hence

$$\begin{aligned} \mathbb{E}[\hat{\beta}] &= \mathbb{E}[(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}] \\ &= (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \underbrace{\mathbb{E}[\mathbf{Y}]}_{=\mathbf{X}\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{X} \beta = \beta. \end{aligned}$$

The LS estimate  $\hat{\beta}$  is hence an **unbiased** estimate for  $\beta$ .

## Variance of the LS Estimate $\hat{\beta}$

It can be shown that the variance of  $\hat{\beta}$  in matrix notation is:

$$\text{Var}(\hat{\beta}) = \sigma^2(\mathbf{X}^T \mathbf{X})^{-1}$$

where

$$\text{Var}(\hat{\beta}) = \begin{bmatrix} \text{Var}(\hat{\beta}_0) & \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) & \text{Cov}(\hat{\beta}_0, \hat{\beta}_2) & \cdots & \text{Cov}(\hat{\beta}_0, \hat{\beta}_p) \\ \text{Cov}(\hat{\beta}_1, \hat{\beta}_0) & \text{Var}(\hat{\beta}_1) & \text{Cov}(\hat{\beta}_1, \hat{\beta}_2) & \cdots & \text{Cov}(\hat{\beta}_1, \hat{\beta}_p) \\ \text{Cov}(\hat{\beta}_2, \hat{\beta}_0) & \text{Cov}(\hat{\beta}_2, \hat{\beta}_1) & \text{Var}(\hat{\beta}_2) & \cdots & \text{Cov}(\hat{\beta}_2, \hat{\beta}_p) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(\hat{\beta}_p, \hat{\beta}_0) & \text{Cov}(\hat{\beta}_p, \hat{\beta}_1) & \text{Cov}(\hat{\beta}_p, \hat{\beta}_2) & \cdots & \text{Var}(\hat{\beta}_p) \end{bmatrix}.$$

and

$$(\mathbf{X}^T \mathbf{X})^{-1} = \begin{bmatrix} n & \sum x_{i1} & \sum x_{i2} & \cdots & \sum x_{ip} \\ \sum x_{i1} & \sum x_{i1}^2 & \sum x_{i1}x_{i2} & \cdots & \sum x_{i1}x_{ip} \\ \sum x_{i2} & \sum x_{i2}x_{i1} & \sum x_{i2}^2 & \cdots & \sum x_{i2}x_{ip} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sum x_{ip} & \sum x_{ip}x_{i1} & \sum x_{ip}x_{i2} & \cdots & \sum x_{ip}^2 \end{bmatrix}^{-1}.$$

## Variance of the LS Estimates for SLR

For SLR,  $(\mathbf{X}^T \mathbf{X})^{-1}$  equals

$$\begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix}^{-1} = \begin{bmatrix} \frac{1}{n} + \frac{\bar{x}^2}{\sum (x_i - \bar{x})^2} & -\frac{\bar{x}}{\sum (x_i - \bar{x})^2} \\ -\frac{\bar{x}}{\sum (x_i - \bar{x})^2} & \frac{1}{\sum (x_i - \bar{x})^2} \end{bmatrix}$$

Hence,

$$\begin{bmatrix} \text{Var}(\hat{\beta}_0) & \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) \\ \text{Cov}(\hat{\beta}_1, \hat{\beta}_0) & \text{Var}(\hat{\beta}_1) \end{bmatrix} = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}.$$

We obtain

$$\text{Var}(\hat{\beta}_0) = \sigma^2 \left( \frac{1}{n} + \frac{\bar{x}^2}{\sum (x_i - \bar{x})^2} \right), \quad \text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum (x_i - \bar{x})^2}.$$

$$\text{Cov}(\hat{\beta}_1, \hat{\beta}_0) = -\frac{\sigma^2 \bar{x}}{\sum (x_i - \bar{x})^2}.$$

## Standard Error (S.E.) of the LS Estimate

For MLR in general,

- $\text{Var}(\hat{\beta}_j) = \sigma^2 \times (\text{the } j\text{th diagonal element of } (\mathbf{X}^T \mathbf{X})^{-1}), j = 0, 1, \dots, p.$
- $\text{Cov}(\hat{\beta}_j, \hat{\beta}_k) = \sigma^2 \times (\text{the } (j, k) \text{ entry of } (\mathbf{X}^T \mathbf{X})^{-1}), j, k = 0, 1, \dots, p.$
- $\text{SE}(\hat{\beta}_j) = \sqrt{\text{Var}(\hat{\beta}_j)}$  but the unknown  $\sigma^2$  is replaced by MSE.

For SLR:

$$\text{SE}(\hat{\beta}_1) = \sqrt{\frac{\text{MSE}}{\sum_{i=1}^n (x_i - \bar{x})^2}}, \quad \text{SE}(\hat{\beta}_0) = \sqrt{\text{MSE}} \cdot \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}.$$

## $t$ -test of a Single $\beta_j$

For an MLR model  $Y_i = \beta_0 + \beta_1 X_{i1} + \dots + \beta_p X_{ip} + \varepsilon_i$ , the  $t$ -statistic for testing  $H_0: \beta_j = \beta_j^0$  is:

$$t = \frac{\hat{\beta}_j - \beta_j^0}{\text{SE.}(\hat{\beta}_j)},$$

which has a  $t$ -distribution with degrees of freedom  $df = n - p - 1$ .

The P-value can be calculated using `pt()` in R based on the alternative hypothesis  $H_1$ :

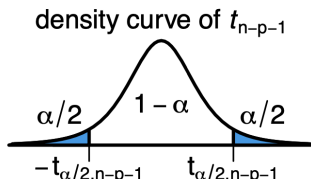
$$\text{P-value} = \begin{cases} \text{pt}(t, \text{df} = n - p - 1) & \text{if } H_1 : \beta_j < \beta_j^0 \\ \text{pt}(t, \text{df} = n - p - 1, \text{lower.tail} = \text{FALSE}) & \text{if } H_1 : \beta_j > \beta_j^0 \\ 2 \cdot \text{pt}(\text{abs}(t), \text{df} = n - p - 1, \text{lower.tail} = \text{FALSE}) & \text{if } H_1 : \beta_j \neq \beta_j^0 \end{cases}$$

## Confidence Intervals for Coefficients

The  $100(1 - \alpha)\%$  confidence interval for  $\beta_j$  is:

$$\hat{\beta}_j \pm t_{(n-p-1, \alpha/2)} \times \text{SE}(\hat{\beta}_j)$$

where  $t_{(n-p-1, \alpha/2)} = t^*$  is the critical value for the  $t_{n-p-1}$  distribution at confidence level  $1 - \alpha$ , i.e.,



This value can be found using either of the following R commands:

```
qt(alpha / 2, df = n - p - 1, lower.tail = FALSE)
qt(1 - alpha / 2, df = n - p - 1)
```

## Multiple $R$ -Squared

**Multiple  $R^2$** , also called the **coefficient of determination**, is defined as:

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}.$$

= proportion of variability in  $Y$  explained by  $X_1, \dots, X_p$ .

- $0 \leq R^2 \leq 1$
- For SLR,  $R^2 = r_{xy}^2$  is the square of the correlation between  $X$  and  $Y$ . So multiple  $R^2$  is a generalization of the correlation.
- For MLR,  $R^2$  is the square of the correlation between  $Y$  and  $\hat{Y}$
- When more terms are added into a model,  $R^2$  may increase or stay the same but never decrease.

## Adjusted $R$ -Squared

Since  $R^2$  always increases as we add terms to the model, some people prefer to use an **adjusted**  $R^2$  defined as:

$$R_{\text{adj}}^2 = 1 - \frac{\text{SSE}/\text{df}_E}{\text{SST}/\text{df}_T} = 1 - \frac{\text{SSE}/(n-p-1)}{\text{SST}/(n-1)} = 1 - \frac{n-1}{n-p-1}(1-R^2).$$

- $-\frac{p}{n-p-1} \leq R_{\text{adj}}^2 \leq R^2 \leq 1$ .
- Unlike  $R^2$ ,  $R_{\text{adj}}^2$  can be negative.
- $R_{\text{adj}}^2$  does not always increase as more variables are added.  
In fact, if unnecessary terms are added,  $R_{\text{adj}}^2$  may decrease.



## Nested Models

**We say Model 1 is nested in Model 2** if Model 1 is a special case of Model 2 (and hence Model 2 is an extension of Model 1).

E.g., for the 4 models below,

$$\text{Model A : } Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$$

$$\text{Model B : } Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

$$\text{Model C : } Y = \beta_0 + \beta_1 X_1 + \beta_3 X_3 + \varepsilon$$

$$\text{Model D : } Y = \beta_0 + \beta_1 (X_1 + X_2) + \varepsilon$$

- B is nested in A ..... since A reduces to B when  $\beta_3 = 0$
- C is also nested in A ..... since A reduces to C when  $\beta_2 = 0$
- D is nested in B ..... since B reduces to D when  $\beta_1 = \beta_2$
- B and C are NOT nested in either way
- D is NOT nested in C

## Nesting Relationship is Transitive

If Model 1 is nested in Model 2, and Model 2 is nested in Model 3, then Model 1 is also nested in Model 3.

For example, for models in the previous slide,

D is nested in B, and B is nested in A,

implies D is also nested in A, which is clearly true because Model A reduces to Model D when

$$\beta_1 = \beta_2, \quad \text{and} \quad \beta_3 = 0.$$

When two models are nested (Model 1 is nested in Model 2),

- the simpler model (Model 1) is called the **reduced model**,
- the more general model (Model 2) is called the **full model**.

## SST of Nested Models

**Question:** Compare the SST's for Model A, B, C, and D. Which one is the largest? Or are they equal?

The 4 models have an identical SST.

$SST = \sum_{i=1}^n (y_i - \bar{y})^2$  only depends on the response  $y$  but not on which predictors are included in the model.

## SSE of Nested Models

When a reduced model is nested in a full model, then

- ❶  $SSE_{\text{reduced}} \geq SSE_{\text{full}}$ , and
- ❷  $SSR_{\text{reduced}} \leq SSR_{\text{full}}$ .

**Proof.**

- Observe that  $\min\{a, b, c, d\} \leq \min\{a, b, c\}$  is always true for any numbers  $a, b, c, d$ .
- In general,  $\min S_1 \leq \min S_2$  if  $S_2$  is a subset of  $S_1$  where  $S_1$  and  $S_2$  are two sets of numbers.
- We will prove (i) for:

$$\text{full model: } y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \varepsilon_i,$$

$$\text{reduced model: } y_i = \beta_0 + \beta_1 x_{i1} + \beta_3 x_{i3} + \varepsilon_i.$$

The proofs for other nested models are similar.

$$\begin{aligned} SSE_{\text{full}} &= \min_{\beta_0, \beta_1, \beta_2, \beta_3} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2} - \beta_3 x_{i3})^2 \\ &\leq \min_{\beta_0, \beta_1, \beta_3} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \beta_3 x_{i3})^2 = SSE_{\text{reduced}}. \end{aligned}$$

Part (ii) follows directly from (i), the identity  $SST = SSR + SSE$ , and the fact that all MLR models of the same data have a common SST.

## General Framework for Testing Nested Models

$H_0$  : reduced model is true      vs.       $H_1$  : full model is true

- Since the reduced model is nested in the full model,

$$\text{SSE}_{\text{reduced}} \geq \text{SSE}_{\text{full}}.$$

- **Simplicity or Accuracy?**

- The full model fits the data better (with a smaller SSE) but is more complicated.
- The reduced model doesn't fit as well but is simpler.
- If  $\text{SSE}_{\text{reduced}} \approx \text{SSE}_{\text{full}}$ , one can sacrifice a bit of accuracy in exchange for simplicity.
- If  $\text{SSE}_{\text{reduced}} \gg \text{SSE}_{\text{full}}$ , it would sacrifice too much in accuracy in exchange for simplicity. The full model is preferred.

## How large is the difference in SSE?

- A larger difference  $SSE_{\text{reduced}} - SSE_{\text{full}}$  is stronger evidence against the reduced model.
- The size of the difference depends on:
  - The **complexity** of the models, i.e., the **difference in the number of parameters**:

$$df_E^{\text{reduced}} - df_E^{\text{full}}$$

- The magnitude of noise  $\sigma^2$ : the larger  $\sigma^2$ , the larger the expected difference even under  $H_0$ .
- A reasonable test statistic is:

$$\frac{(SSE_{\text{reduced}} - SSE_{\text{full}})/(df_E^{\text{reduced}} - df_E^{\text{full}})}{\sigma^2}$$

- We estimate the unknown  $\sigma^2$  with MSE.
- Use  $MSE_{\text{full}}$  instead of  $MSE_{\text{reduced}}$  since the full model is always true under  $H_0$ .

# The F-Statistic

$$F = \frac{(\text{SSE}_{\text{reduced}} - \text{SSE}_{\text{full}}) / (df_E^{\text{reduced}} - df_E^{\text{full}})}{\text{MSE}_{\text{full}}}.$$

- $df_E^{\text{reduced}}$ : degrees of freedom for SSE of the reduced model,
- $df_E^{\text{full}}$ : degrees of freedom for SSE of the full model,
- $F \geq 0$  since  $\text{SSE}_{\text{reduced}} \geq \text{SSE}_{\text{full}}$ ,
- The smaller the  $F$ -statistic, the more the reduced model is favored,
- Under  $H_0$ , the  $F$ -statistic follows an  $F$ -distribution with:

$$df_1 = df_E^{\text{reduced}} - df_E^{\text{full}}, \quad df_2 = df_E^{\text{full}}.$$

# Testing overall significance of the model

## Testing the hypotheses

$$H_0 : \beta_1 = \cdots = \beta_p = 0 \quad \text{v.s.} \quad H_a : \text{not all } \beta_1, \dots, \beta_p = 0$$

This is a test to evaluate the **overall significance** of a model.

- Full:  $y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \varepsilon_i$ ,
- Reduced:  $y_i = \beta_0 + \varepsilon_i$  (all predictors are unnecessary),
- The LS estimate for  $\beta_0$  in the reduced model is  $\hat{\beta}_0 = \bar{y}$ , so

$$\text{SSE}_{\text{reduced}} = \sum_{i=1}^n (y_i - \hat{\beta}_0)^2 = \sum_{i=1}^n (y_i - \bar{y})^2 = \text{SST}_{\text{full}},$$

- $\text{df}_E^{\text{full}} = n - p - 1$ ,
- $\text{df}_E^{\text{reduced}} = n - 1$  since the reduced model has 0 predictors.



## Testing overall significance of the model

$$\begin{aligned}
 F &= \frac{(\text{SSE}_{\text{reduced}} - \text{SSE}_{\text{full}}) / (df_E^{\text{reduced}} - df_E^{\text{full}})}{\text{MSE}_{\text{full}}} \\
 &= \frac{(\text{SST}_{\text{full}} - \text{SSE}_{\text{full}}) / [n - 1 - (n - p - 1)]}{\text{SSE}_{\text{full}} / (n - p - 1)} \\
 &= \frac{\text{SSR}_{\text{full}} / p}{\text{SSE}_{\text{full}} / (n - p - 1)} = \frac{\text{MSR}_{\text{full}}}{\text{MSE}_{\text{full}}}.
 \end{aligned}$$

Moreover,  $F \sim F_{p, n-p-1}$  under  $H_0 : \beta_1 = \beta_2 = \dots = \beta_p = 0$ .

**Note:** In R, the  $F$  statistic and  $p$ -value are displayed in the last line of the output of the `summary()` command.

## $F$ -Test on a Single $\beta_j$ is Equivalent to $t$ -Test

Suppose that we want to test a single  $\beta_3 = 0$  in the model:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon.$$

- One can do a  $t$ -test by reading the  $t$ -statistic and  $p$ -value for  $X_3$  from the output of `summary(lm(Y ~ X1 + X2 + X3))`.
- Alternatively, one can conduct an  $F$ -test comparing the models:
  - **Full model:**  $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$ ,
  - **Reduced model:**  $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$

by using the following R command

```
anova(lm(Y ~ X1 + X2 + X3), lm(Y ~ X1 + X2))
```

# Residual Analysis

- **Residual analysis** is used to check the assumptions of linear regression models.
- The assumptions include:

- ① **Linearity**: the relationship between  $X$  and  $Y$  is linear, i.e.:

$$\mathbb{E}[Y|X = \mathbf{x}] = \mathbf{x}^T \theta.$$

- ② **Homoscedasticity**: the variance of the response variable  $Y$  is constant for all values of the independent variable  $X$ , i.e.:

$$\text{Var}(Y|X = \mathbf{x}) = \sigma^2.$$

- ③ **Independence**: the observations of the response variable  $Y$  are independent.
- ④ **Normality**: the conditional distribution of  $Y$  given  $X = \mathbf{x}$  is normally distributed, i.e.:

$$Y|X = \mathbf{x} \sim \mathcal{N}(\mathbf{x}^T \beta, \sigma^2).$$

- These assumptions are usually examined using the residuals:

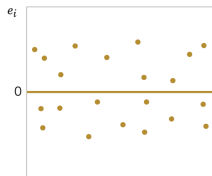
$$e_i = y_i - \hat{y}_i, \quad i = 1, \dots, n$$

where  $\hat{y}_i = \mathbf{x}^T \beta$ .

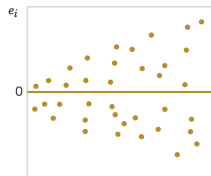
## Residual Plots

- Plot the residuals as points  $(\hat{y}_i, e_i)$ , or  $(x_i, e_i)$  for each observation  $i = 1, \dots, n$ .
- If assumptions 1–3 hold, the residuals should be randomly scattered on the  $x$ - $y$  plane around the line  $y = 0$ .
- If one of the assumptions is violated, e.g., non-linearity or heteroscedasticity, the residuals will show a specific pattern.
- Residual plots also help detect the presence of **outliers**.
- To assess the normality assumption (Assumption 4), we commonly use the **Normal Q-Q Plot**.

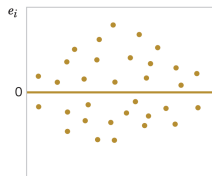
# Residual Analysis



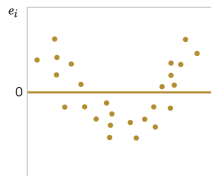
(a)



(b)



(c)

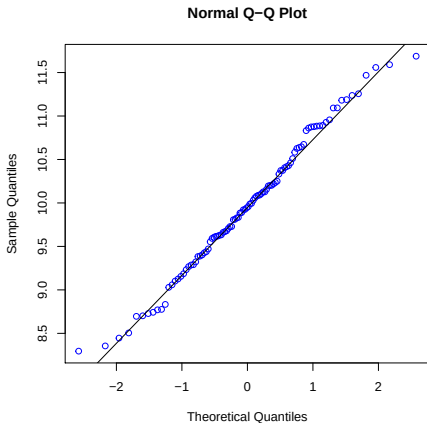


(d)

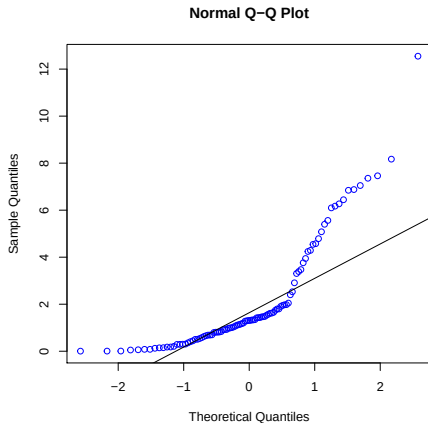
- (a) Model assumptions are satisfied.
- (b) Variance increases over time or with the magnitude of  $x_i$  or  $y_i$ .
- (c) Heteroscedasticity: variance is not constant.
- (d) The relationship between  $X$  and  $Y$  is non-linear.

# Residual Analysis

- Checking the normality assumption using the **Normal Q-Q Plot**.



Data follows normal distribution



Data does not follow normal distribution